

Solve Any Data Analysis Problem

Chapter 5 - Project 4 - Example solution - Part 1

We need to find and extract:

- annual admissions breakdown
- breakdown of genre
- breakdown of distributors

Figure out where relevant tables are for a few years (2018 onwards so we have both pre- and "post-" COVID periods)

```
In [1]: import numpy as np
import pandas as pd

import pdfplumber

from IPython.display import display
```

Let's try a sample snippet to extract tables from a specific page

```
In [2]: pdf_path = "./files/2019 - BFI yearbook 2019 - 888.pdf"
page_num = 11

page_tables = []

with pdfplumber.open(pdf_path) as pdf:
    page = pdf.pages[page_num-1]
    # page_tables is a list of lists of Lists(!)
    # extracted from special Table objects
    page_tables = [t.extract() for t in page.find_tables()]

page_tables
```

```
Out[2]: [[['Month', '2017 (million)', '2018 (million)', '% +/- on 2017'],
['January', '15.0', '16.2', '8.0'],
['February', '16.5', '16.1', '-2.3'],
['March', '16.2', '13.5', '-16.2'],
['April', '15.6', '15.5', '-0.9'],
['May', '11.3', '13.7', '21.0'],
['June', '9.6', '10.4', '8.6'],
['July', '17.8', '15.6', '-12.3'],
['August', '14.5', '19.2', '32.8'],
['September', '10.8', '10.1', '-6.0'],
['October', '12.1', '16.0', '32.6'],
['November', '14.1', '14.8', '5.4'],
['December', '17.2', '15.7', '-8.7'],
['Total', '170.6', '177.0', '3.7']]]
```

Convert to a DataFrame

```
In [3]: table = page_tables[0]

pd.DataFrame(table[1:-1], columns=table[0])
```

```
Out[3]:
```

	Month	2017 (million)	2018 (million)	% +/- on 2017
0	January	15.0	16.2	8.0
1	February	16.5	16.1	-2.3
2	March	16.2	13.5	-16.2
3	April	15.6	15.5	-0.9
4	May	11.3	13.7	21.0
5	June	9.6	10.4	8.6
6	July	17.8	15.6	-12.3
7	August	14.5	19.2	32.8
8	September	10.8	10.1	-6.0
9	October	12.1	16.0	32.6
10	November	14.1	14.8	5.4
11	December	17.2	15.7	-8.7

Let's now create a reusable function to extract and print all the tables in a PDF

```
In [4]: def extract_tables(pdf_path, pages=[], print_tables=True):
        """
        Extract all tables found in a PDF.

        `pdf_path`: file path pointing to the PDF
        `pages`: the page number(s) to read
        `print_tables`: whether to also print out all the tables that are found (default)

        returns: a list of pandas DataFrames
        """

        print(f"Reading {pdf_path}")

        tables = []

        with pdfplumber.open(pdf_path) as pdf:
            for page_num in pages:
                page = pdf.pages[page_num-1]

                # page_tables is a list of lists of lists(!)
                # extracted from special Table objects
                page_tables = [t.extract() for t in page.find_tables()]
```

```

        # in each case, table is now a list of lists
        # first list is column headers
        dfs = [pd.DataFrame(table[1:-1], columns=table[0]) for table in page_tables]

        tables.extend(dfs)

    print(f"{len(tables)} tables found.")

    if len(tables) > 0:
        if print_tables:
            for index, df in enumerate(tables):
                print(f"\n#####\n\tTable {index}\n#####")
                display(df)

    return tables

```

Test our function

```
In [5]: tables = extract_tables("./files/2019 - BFI yearbook 2019 - 888.pdf", pages=[11,34,
```

Reading ./files/2019 - BFI yearbook 2019 - 888.pdf
3 tables found.

```
#####
      Table 0
#####
```

	Month	2017 (million)	2018 (million)	% +/- on 2017
0	January	15.0	16.2	8.0
1	February	16.5	16.1	-2.3
2	March	16.2	13.5	-16.2
3	April	15.6	15.5	-0.9
4	May	11.3	13.7	21.0
5	June	9.6	10.4	8.6
6	July	17.8	15.6	-12.3
7	August	14.5	19.2	32.8
8	September	10.8	10.1	-6.0
9	October	12.1	16.0	32.6
10	November	14.1	14.8	5.4
11	December	17.2	15.7	-8.7

```
#####
      Table 1
#####
```

	Genre	Number of\nreleases	% of\nreleases	Gross box\noffice\n(£ million)	% of total\nbox office	Top performing title
0	Action	77	9.8	361.3	27.7	Avengers: Infinity War
1	Animation	46	5.8	242.2	18.6	Incredibles 2
2	Drama	246	31.3	154.1	11.8	A Star Is Born
3	Comedy	144	18.3	88.3	6.8	Johnny English Strikes Again
4	Biopic	5	0.6	78.5	6.0	Bohemian Rhapsody
5	Musical	5	0.6	67.0	5.1	Mamma Mia! Here We Go Again
6	Family	10	1.3	60.3	4.6	Mary Poppins Returns
7	Fantasy	4	0.5	57.6	4.4	Fantastic Beasts: The Crimes of Grindelwald
8	Horror	38	4.8	57.1	4.4	A Quiet Place
9	Adventure	10	1.3	36.4	2.8	Ready Player One
10	Sci-fi	14	1.8	30.8	2.4	Solo: A Star Wars Story
11	Thriller	42	5.3	30.6	2.3	The Equalizer 2
12	Crime	10	1.3	17.9	1.4	Widows
13	Romance	17	2.2	11.8	0.9	The Guernsey Literary and Potato Peel Pie Society
14	Documentary	112	14.2	9.3	0.7	Free Solo
15	War	3	0.4	1.3	0.1	Journey's End
16	Western	2	0.3	0.1	<0.1	Sweet Country
17	Mystery	2	0.3	0.1	<0.1	Dark River

Table 2
#####

	Distributor	Market share\n(%)	Films on release\nnin 2018	Box office gross\n(£ million)
0	Walt Disney	23.6	24	325.6
1	Universal	19.5	40	268.5
2	20th Century Fox	14.5	28	199.3
3	Warner Bros	13.9	31	191.4
4	Sony	10.7	33	146.8
5	Paramount	4.8	12	66.0
6	eOne Films	3.2	22	43.9
7	StudioCanal	2.8	31	38.4
8	Lionsgate	1.5	21	21.2
9	Entertainment	1.1	9	15.6
10	Top 10 total	95.5	251	1,316.8
11	Others (128 distributors)	4.5	715	61.4

2018

For 2018 all necessary tables are in the same PDF so let's extract them all here.

```
In [6]: tables_2018 = extract_tables("./files/2019 - BFI yearbook 2019 - 888.pdf", pages=[1
```

```
Reading ./files/2019 - BFI yearbook 2019 - 888.pdf
3 tables found.
```

```
#####
      Table 0
#####
```

	Month	2017 (million)	2018 (million)	% +/- on 2017
0	January	15.0	16.2	8.0
1	February	16.5	16.1	-2.3
2	March	16.2	13.5	-16.2
3	April	15.6	15.5	-0.9
4	May	11.3	13.7	21.0
5	June	9.6	10.4	8.6
6	July	17.8	15.6	-12.3
7	August	14.5	19.2	32.8
8	September	10.8	10.1	-6.0
9	October	12.1	16.0	32.6
10	November	14.1	14.8	5.4
11	December	17.2	15.7	-8.7

Table 1
#####

	Genre	Number of\nreleases	% of\nreleases	Gross box\noffice\n(£ million)	% of total\nbox office	Top performing title
0	Action	77	9.8	361.3	27.7	Avengers: Infinity War
1	Animation	46	5.8	242.2	18.6	Incredibles 2
2	Drama	246	31.3	154.1	11.8	A Star Is Born
3	Comedy	144	18.3	88.3	6.8	Johnny English Strikes Again
4	Biopic	5	0.6	78.5	6.0	Bohemian Rhapsody
5	Musical	5	0.6	67.0	5.1	Mamma Mia! Here We Go Again
6	Family	10	1.3	60.3	4.6	Mary Poppins Returns
7	Fantasy	4	0.5	57.6	4.4	Fantastic Beasts: The Crimes of Grindelwald
8	Horror	38	4.8	57.1	4.4	A Quiet Place
9	Adventure	10	1.3	36.4	2.8	Ready Player One
10	Sci-fi	14	1.8	30.8	2.4	Solo: A Star Wars Story
11	Thriller	42	5.3	30.6	2.3	The Equalizer 2
12	Crime	10	1.3	17.9	1.4	Widows
13	Romance	17	2.2	11.8	0.9	The Guernsey Literary and Potato Peel Pie Society
14	Documentary	112	14.2	9.3	0.7	Free Solo
15	War	3	0.4	1.3	0.1	Journey's End
16	Western	2	0.3	0.1	<0.1	Sweet Country
17	Mystery	2	0.3	0.1	<0.1	Dark River

Table 2
#####

	Distributor	Market share\n(%)	Films on release\nnin 2018	Box office gross\nn(£ million)
0	Walt Disney	23.6	24	325.6
1	Universal	19.5	40	268.5
2	20th Century Fox	14.5	28	199.3
3	Warner Bros	13.9	31	191.4
4	Sony	10.7	33	146.8
5	Paramount	4.8	12	66.0
6	eOne Films	3.2	22	43.9
7	StudioCanal	2.8	31	38.4
8	Lionsgate	1.5	21	21.2
9	Entertainment	1.1	9	15.6
10	Top 10 total	95.5	251	1,316.8
11	Others (128 distributors)	4.5	715	61.4

The first table is the monthly breakdown we're looking for.

```
In [7]: admissions_2018 = (
        tables_2018[0]
        .iloc[:,[0, 2]] # keep month and 2018 admissions column
    )

admissions_2018.columns = ["Month", "Admissions (million)"]

admissions_2018.insert(0, "Year", 2018)

admissions_2018.head()
```

```
Out[7]:
```

	Year	Month	Admissions (million)
0	2018	January	16.2
1	2018	February	16.1
2	2018	March	13.5
3	2018	April	15.5
4	2018	May	13.7

```
In [8]: admissions_2018.tail()
```


Out[8]:

	Year	Month	Admissions (million)
7	2018	August	19.2
8	2018	September	10.1
9	2018	October	16.0
10	2018	November	14.8
11	2018	December	15.7

Genres (2018)

In [9]: `tables_2018[1]`

Out[9]:

	Genre	Number of\nreleases	% of\nreleases	Gross box\noffice\n(£ million)	% of total\nbox office	Top performing title
0	Action	77	9.8	361.3	27.7	Avengers: Infinity War
1	Animation	46	5.8	242.2	18.6	Incredibles 2
2	Drama	246	31.3	154.1	11.8	A Star Is Born
3	Comedy	144	18.3	88.3	6.8	Johnny English Strikes Again
4	Biopic	5	0.6	78.5	6.0	Bohemian Rhapsody
5	Musical	5	0.6	67.0	5.1	Mamma Mia! Here We Go Again
6	Family	10	1.3	60.3	4.6	Mary Poppins Returns
7	Fantasy	4	0.5	57.6	4.4	Fantastic Beasts: The Crimes of Grindelwald
8	Horror	38	4.8	57.1	4.4	A Quiet Place
9	Adventure	10	1.3	36.4	2.8	Ready Player One
10	Sci-fi	14	1.8	30.8	2.4	Solo: A Star Wars Story
11	Thriller	42	5.3	30.6	2.3	The Equalizer 2
12	Crime	10	1.3	17.9	1.4	Widows
13	Romance	17	2.2	11.8	0.9	The Guernsey Literary and Potato Peel Pie Society
14	Documentary	112	14.2	9.3	0.7	Free Solo
15	War	3	0.4	1.3	0.1	Journey's End
16	Western	2	0.3	0.1	<0.1	Sweet Country
17	Mystery	2	0.3	0.1	<0.1	Dark River

```
In [10]: genres_2018 = (
    tables_2018[1]
    .drop(columns=[tables_2018[1].columns[2], tables_2018[1].columns[4]])
```

```
)

genres_2018.insert(0, "Year", 2018)

genres_2018.columns = ["Year", "Genre", "Number of releases", "Gross box office (£
genres_2018.head()
```

Out[10]:

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
0	2018	Action	77	361.3	Avengers: Infinity War
1	2018	Animation	46	242.2	Incredibles 2
2	2018	Drama	246	154.1	A Star Is Born
3	2018	Comedy	144	88.3	Johnny English Strikes Again
4	2018	Biopic	5	78.5	Bohemian Rhapsody

In [11]:

```
genres_2018.tail()
```

Out[11]:

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
13	2018	Romance	17	11.8	The Guernsey Literary and Potato Peel Pie Society
14	2018	Documentary	112	9.3	Free Solo
15	2018	War	3	1.3	Journey's End
16	2018	Western	2	0.1	Sweet Country
17	2018	Mystery	2	0.1	Dark River

Distributors (2018)

In [12]:

```
tables_2018[2]
```

Out[12]:

	Distributor	Market share\n(%)	Films on release\nnin 2018	Box office gross\n(£ million)
0	Walt Disney	23.6	24	325.6
1	Universal	19.5	40	268.5
2	20th Century Fox	14.5	28	199.3
3	Warner Bros	13.9	31	191.4
4	Sony	10.7	33	146.8
5	Paramount	4.8	12	66.0
6	eOne Films	3.2	22	43.9
7	StudioCanal	2.8	31	38.4
8	Lionsgate	1.5	21	21.2
9	Entertainment	1.1	9	15.6
10	Top 10 total	95.5	251	1,316.8
11	Others (128 distributors)	4.5	715	61.4

```
In [13]: distributors_2018 = (
    tables_2018[2]
    .drop(index=[10]) # drop rows relating to totals EXCEPT the row totalling "othe
    )

    distributors_2018.insert(0, "Year", 2018)

    distributors_2018.columns = ["Year", "Distributor", "Market share", "Films on relea
    distributors_2018.head()
```

Out[13]:

	Year	Distributor	Market share	Films on release	Box office gross (£ million)
0	2018	Walt Disney	23.6	24	325.6
1	2018	Universal	19.5	40	268.5
2	2018	20th Century Fox	14.5	28	199.3
3	2018	Warner Bros	13.9	31	191.4
4	2018	Sony	10.7	33	146.8

```
In [14]: distributors_2018.tail()
```

Out[14]:

	Year	Distributor	Market share	Films on release	Box office gross (£ million)
6	2018	eOne Films	3.2	22	43.9
7	2018	StudioCanal	2.8	31	38.4
8	2018	Lionsgate	1.5	21	21.2
9	2018	Entertainment	1.1	9	15.6
11	2018	Others (128 distributors)	4.5	715	61.4

2019

In [15]: `tables_2019 = extract_tables("./files/2020 - BFI Statistical Yearbook 2020 - 12815.`

Reading ./files/2020 - BFI Statistical Yearbook 2020 - 12815.pdf
4 tables found.

```
#####  
      Table 0  
#####
```

	Territory	Admissions 2018\n(million)	Admissions 2019\n(million)	+/- 2018 (%)
0	China	1,717	1,727	0.6
1	India	1,463	1,592	8.8
2	USA and Canada	1,310	1,256	-4.1
3	Mexico	332	350	5.5
4	South Korea	216	227	4.8
5	Russia	200	219	9.5
6	France	201	213	6.0
7	Japan	169	195	15.2
8	UK	177	176	-0.5
9	Brazil	163	172	5.6
10	Indonesia	135	146	7.8
11	Germany	105	119	12.6
12	Italy	92	105	14.2
13	Spain	99	105	4.8
14	Poland	60	61	1.5

#####

Table 1

#####

	Month	2018 (million)	2019 (million)	% +/- on 2018
0	January	16.2	13.7	-15.4
1	February	16.1	12.2	-24.0
2	March	13.5	11.4	-15.5
3	April	15.5	16.0	3.1
4	May	13.7	16.6	20.8
5	June	10.4	13.9	33.7
6	July	15.6	18.7	19.5
7	August	19.2	15.6	-19.1
8	September	10.1	11.0	8.2
9	October	16.1	16.3	1.8
10	November	14.9	12.2	-17.8
11	December	15.7	18.5	18.1

Table 2
#####

	Genre	Number of\nreleases	% of\nreleases	Gross\nbox office\n(£ million)	% of total\nbox office	Top performing title
0	Action	102	13.4	328.3	25.2	Avengers: Endgame
1	Animation	40	5.2	312.1	24.0	The Lion King
2	Drama	264	34.6	166.4	12.8	Downton Abbey
3	Comedy	136	17.8	83.5	6.4	Yesterday
4	Biopic	19	2.5	80.7	6.2	Rocketman
5	Thriller	20	2.6	75.3	5.8	Joker
6	Sci-fi	3	0.4	64.3	4.9	Star Wars: The Rise of Skywalker
7	Horror	37	4.8	60.6	4.7	It Chapter Two
8	Adventure	8	1.0	44.5	3.4	Aladdin
9	Family	5	0.7	42.7	3.3	Dumbo
10	Romance	7	0.9	18.9	1.5	Last Christmas
11	Musical	8	1.0	12.3	0.9	Cats
12	Documentary	99	13.0	7.8	0.6	Apollo 11
13	Crime	11	1.4	2.8	0.2	21 Bridges
14	Mystery	1	0.1	0.5	<0.1	The Souvenir
15	War	2	0.3	0.1	<0.1	Saving Private Ryan (D-Day 75th Anniversary)
16	Fantasy	2	0.3	<0.1	<0.1	Gwen

#####

Table 3

#####

	Distributor	Market share\n(%)	Films on release\nnin 2019	Box office gross\n(£ million)
0	Walt Disney	37.9	24	506.8
1	Universal	13.9	39	185.6
2	Warner Bros	12.4	40	166.3
3	Sony	9.3	27	124.7
4	Paramount	5.9	13	79.0
5	20th Century Fox*	5.7	30	75.5
6	Lionsgate	4.2	19	56.4
7	Entertainment One	3.8	27	50.4
8	STX Entertainment	1.2	7	15.4
9	StudioCanal	1.0	22	14.0
10	Top 10	95.4	248	1,274.1
11	Others (Total 130 distributors)	4.6	671	61.6

Here it looks like admissions is the second table.

Technically we could have extracted the 2018 figures from here as well. If we wanted a genre or distributor-level breakdown for 2018, we would have needed to open those PDFs anyway, so it doesn't make a difference.

If you were being really careful, you could also compare these figures with the ones in the 2018 PDF to make sure they match...

```
In [16]: admissions_2019 = (
    tables_2019[1]
    .iloc[:,[0, 2]] # ignore first two rows, last 2 rows, and keep first and third
)

admissions_2019.columns = ["Month", "Admissions (million)"]
admissions_2019.insert(0, "Year", 2019)

admissions_2019.head()
```

Out[16]:

	Year	Month	Admissions (million)
0	2019	January	13.7
1	2019	February	12.2
2	2019	March	11.4
3	2019	April	16.0
4	2019	May	16.6

In [17]: `admissions_2019.tail()`

Out[17]:

	Year	Month	Admissions (million)
7	2019	August	15.6
8	2019	September	11.0
9	2019	October	16.3
10	2019	November	12.2
11	2019	December	18.5

Genres (2019)

In [18]:

```
genres_2019 = (
    tables_2019[2]
    .drop(columns=[tables_2019[2].columns[2], tables_2019[2].columns[4]])
)

genres_2019.insert(0, "Year", 2019)

genres_2019.columns = ["Year", "Genre", "Number of releases", "Gross box office (£
genres_2019.head(7)
```

Out[18]:

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
0	2019	Action	102	328.3	Avengers: Endgame
1	2019	Animation	40	312.1	The Lion King
2	2019	Drama	264	166.4	Downton Abbey
3	2019	Comedy	136	83.5	Yesterday
4	2019	Biopic	19	80.7	Rocketman
5	2019	Thriller	20	75.3	Joker
6	2019	Sci-fi	3	64.3	Star Wars: The Rise of Skywalker

```
In [19]: genres_2019.tail()
```

```
Out[19]:
```

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
12	2019	Documentary	99	7.8	Apollo 11
13	2019	Crime	11	2.8	21 Bridges
14	2019	Mystery	1	0.5	The Souvenir
15	2019	War	2	0.1	Saving Private Ryan (D-Day 75th Anniversary)
16	2019	Fantasy	2	<0.1	Gwen

This code is very similar to the one for 2018. We *could* write a generic function to extract and transform a genre-level breakdown from any PDF, but we cannot assume they will all take the same form.

We generally don't want to assume data across multiple files will have the same format, but this is particularly true for PDF reports spanning many years.

Distributors (2019)

```
In [20]: distributors_2019 = (
    tables_2019[3]
    .drop(index=[10]) # drop totals EXCEPT the row totalling "others"
)

distributors_2019.insert(0, "Year", 2019)

distributors_2019.columns = ["Year", "Distributor", "Market share", "Films on release", "Box office gross (£ million)"]

distributors_2019.head()
```

```
Out[20]:
```

	Year	Distributor	Market share	Films on release	Box office gross (£ million)
0	2019	Walt Disney	37.9	24	506.8
1	2019	Universal	13.9	39	185.6
2	2019	Warner Bros	12.4	40	166.3
3	2019	Sony	9.3	27	124.7
4	2019	Paramount	5.9	13	79.0

```
In [21]: distributors_2019.tail()
```

Out[21]:

	Year	Distributor	Market share	Films on release	Box office gross (£ million)
6	2019	Lionsgate	4.2	19	56.4
7	2019	Entertainment One	3.8	27	50.4
8	2019	STX Entertainment	1.2	7	15.4
9	2019	StudioCanal	1.0	22	14.0
11	2019	Others (Total 130 distributors)	4.6	671	61.6

2020

In [22]: `tables_2020 = extract_tables("./files/2021 - BFI Statistical Yearbook 2021 - 24979.pdf")`

Reading ./files/2021 - BFI Statistical Yearbook 2021 - 24979.pdf
4 tables found.

```
#####  
      Table 0  
#####
```

	Admissions\n(million)	2016	2017	2018	2019	2020	% change\nnon 2019
0	January	14.0	15.0	16.2	13.7	16.5	20.4
1	February	15.4	16.5	16.1	12.2	14.5	19.0
2	March	13.4	16.2	13.5	11.4	4.8	-57.9
3	April	13.1	15.6	15.5	16.0	-	-100.0
4	May	12.5	11.3	13.7	16.6	-	-100.0
5	June	10.7	9.6	10.4	13.9	-	-100.0
6	July	16.6	17.8	15.6	18.7	0.4	-97.9
7	August	18.1	14.5	19.2	15.6	2.1	-86.7
8	September	11.7	10.8	10.1	11.0	2.7	-75.8
9	October	15.2	12.1	16.1	16.3	2.2	-86.6
10	November	12.4	14.1	14.9	12.2	0.3	-97.5
11	December	15.2	17.2	15.7	18.5	0.5	-97.2

```
#####  
      Table 1  
#####
```

	Genre	Number of\nreleases	% of\nreleases	Gross box\noffice\n(£ million)	% of total\nbox office	Average number\nof sites at widest\npoint of release	per
0	Action	36	9.4%	48.5	19.7%	167	
1	War	3	0.8%	44.3	17.9%	296	
2	Animation	19	5.0%	34.8	14.1%	260	S H
3	Thriller	15	3.9%	32.8	13.3%	192	
4	Drama	140	36.7%	31.4	12.7%	83	c Co
5	Comedy	65	17.1%	24.0	9.7%	108	Jo
6	Crime	8	2.1%	12.5	5.1%	118	Ge
7	Horror	24	6.3%	12.1	4.9%	172	The
8	Adventure	9	2.4%	3.8	1.6%	152	Th
9	Fantasy	2	0.5%	0.9	0.4%	315	F
10	Romance	6	1.6%	0.4	0.2%	30	H
11	Documentary	42	11.0%	0.4	0.2%	19	I /
12	Western	2	0.5%	0.3	0.1%	53	Tru Of
13	Family	3	0.8%	0.3	0.1%	85	The C
14	Biopic	3	0.8%	0.2	0.1%	16	M
15	Sci fi	4	1.0%	0.1	0.0%	18	C

Table 2
#####

	Distributor	Market share\nin 2020\n(%)	Number of films\nnon release\nin 2020	Box office\ngross in 2020\n(£ million)	Number of films\nnon release\nin 2019
0	Entertainment One	15.9	30	49.4	28
1	Sony	15.6	38	48.4	39
2	Walt Disney	14.3	58	44.3	47
3	Universal	12.5	73	39.0	55
4	Warner Bros	11.7	89	36.5	64
5	Paramount	8.0	22	25.0	13
6	Lionsgate	4.5	29	14.0	23
7	StudioCanal	4.5	46	13.9	31
8	Entertainment Film Distributors	4.3	6	13.3	10
9	Shear Entertainment*	1.3	1	4.0	n/a
10	Top 10	92.5	392	287.8	310
11	Other distributors (143)	7.5	534	23.2	844

Table 3
#####

	Distributor	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
0	Entertainment One	5.1	6.7	9.0	8.0	3.9	8.4	2.5	3.2	3.8	15.9
1	Sony	7.2	18.0	8.7	6.2	11.8	6.6	10.3	10.7	9.3	15.6
2	Walt Disney	8.7	10.2	15.2	10.1	20.0	23.2	19.7	23.6	37.9	14.3
3	Universal	11.8	10.7	15.1	11.2	21.6	14.0	16.0	19.5	13.9	12.5
4	Warner Bros	18.2	12.9	17.2	15.9	9.0	15.6	16.6	13.9	12.4	11.7
5	Paramount	16.3	7.7	7.8	5.8	4.0	5.4	3.7	4.8	5.9	8.0
6	Lionsgate	-	5.7	4.7	5.5	4.0	4.0	6.3	1.5	4.2	4.5
7	Optimum/StudioCanal1	3.8	-	2.8	6.7	4.7	1.5	4.2	2.8	1.0	4.5
8	Entertainment Film Distributors	6.7	3.1	1.9	5.2	1.6	1.5	1.7	1.1	-	4.3
9	Shear Entertainment	-	-	-	-	-	-	-	-	-	1.3
10	20th Century Fox2	12.1	16.1	13.1	21.8	14.9	15.7	15.1	14.5	5.7	-
11	STX Entertainment	-	-	-	-	-	-	-	-	1.2	-
12	Momentum3	4.6	4.3	-	-	-	-	-	-	-	-
13	Top 10 total4	94.4	95.4	95.5	96.3	95.5	95.9	96.1	95.5	95.4	92.5
14	Others	5.6	4.6	4.5	3.7	4.5	4.1	3.9	4.5	4.6	7.5

```
In [23]: admissions_2020 = (
            tables_2020[0]
            .iloc[:,[0,5]]
        )

admissions_2020.columns = ["Month", "Admissions (million)"]
admissions_2020.insert(0, "Year", 2020)

admissions_2020.head()
```

```
Out[23]:
```

	Year	Month	Admissions (million)
0	2020	January	16.5
1	2020	February	14.5
2	2020	March	4.8
3	2020	April	-
4	2020	May	-

Let's replace those hyphens with explicit missing (NaN) values

```
In [24]: admissions_2020 = admissions_2020.replace("-", np.NaN)
admissions_2020.head()
```

Out[24]:

	Year	Month	Admissions (million)
0	2020	January	16.5
1	2020	February	14.5
2	2020	March	4.8
3	2020	April	NaN
4	2020	May	NaN

In [25]: `admissions_2020.tail()`

Out[25]:

	Year	Month	Admissions (million)
7	2020	August	2.1
8	2020	September	2.7
9	2020	October	2.2
10	2020	November	0.3
11	2020	December	0.5

Genres

In [26]:

```
genres_2020 = (
    tables_2020[1]
    .drop(columns=[tables_2020[1].columns[2], tables_2020[1].columns[4], tables_2020[1].columns[5]])
)

genres_2020.insert(0, "Year", 2020)

genres_2020.columns = ["Year", "Genre", "Number of releases", "Gross box office (£m)", "Number of weeks in release"]

genres_2020.head(8)
```


Out[26]:

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
0	2020	Action	36	48.5	Bad Boys For Life
1	2020	War	3	44.3	1917
2	2020	Animation	19	34.8	Sonic The Hedgehog
3	2020	Thriller	15	32.8	Tenet
4	2020	Drama	140	31.4	The Personal History of David Copperfield
5	2020	Comedy	65	24.0	Jojo Rabbit
6	2020	Crime	8	12.5	The Gentlemen
7	2020	Horror	24	12.1	The Invisible Man

In [27]: `genres_2020.tail()`

Out[27]:

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
11	2020	Documentary	42	0.4	I Am Greta
12	2020	Western	2	0.3	True History Of The Kelly Gang
13	2020	Family	3	0.3	Max Winslow And The House Of Secrets
14	2020	Biopic	3	0.2	Mr. Jones
15	2020	Sci fi	4	0.1	Color Out Of Space

Looks like a slight data error where table rows spill over.

Let's manually fix these since we're here.

In [28]: `genres_2020["Top performing title"] = genres_2020["Top performing title"].str.replace("\n", " ")`
`genres_2020.head()`

Out[28]:

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
0	2020	Action	36	48.5	Bad Boys For Life
1	2020	War	3	44.3	1917
2	2020	Animation	19	34.8	Sonic The Hedgehog
3	2020	Thriller	15	32.8	Tenet
4	2020	Drama	140	31.4	The Personal History of David Copperfield

In [29]: `genres_2020.tail()`

Out[29]:

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
11	2020	Documentary	42	0.4	I Am Greta
12	2020	Western	2	0.3	True History Of The Kelly Gang
13	2020	Family	3	0.3	Max Winslow And The House Of Secrets
14	2020	Biopic	3	0.2	Mr. Jones
15	2020	Sci fi	4	0.1	Color Out Of Space

Distributors

In [30]: `tables_2020[2]`

Out[30]:

	Distributor	Market share\nin 2020\n(%)	Number of films\nnon release\nin 2020	Box office\ngross in 2020\n(£ million)	Number of films\nnon release\nin 2019
0	Entertainment One	15.9	30	49.4	28
1	Sony	15.6	38	48.4	39
2	Walt Disney	14.3	58	44.3	47
3	Universal	12.5	73	39.0	55
4	Warner Bros	11.7	89	36.5	64
5	Paramount	8.0	22	25.0	13
6	Lionsgate	4.5	29	14.0	23
7	StudioCanal	4.5	46	13.9	31
8	Entertainment Film Distributors	4.3	6	13.3	10
9	Shear Entertainment*	1.3	1	4.0	n/a
10	Top 10	92.5	392	287.8	310
11	Other distributors (143)	7.5	534	23.2	844

```
In [31]: distributors_2020 = (
    tables_2020[2]
    .iloc[:, :4]
    .drop(index=[10]) # drop totals EXCEPT the row totalling "others"
)

distributors_2020.insert(0, "Year", 2020)

distributors_2020.columns = ["Year", "Distributor", "Market share", "Films on release", "Box office gross (£ million)"]

distributors_2020.head()
```

Out[31]:

	Year	Distributor	Market share	Films on release	Box office gross (£ million)
0	2020	Entertainment One	15.9	30	49.4
1	2020	Sony	15.6	38	48.4
2	2020	Walt Disney	14.3	58	44.3
3	2020	Universal	12.5	73	39.0
4	2020	Warner Bros	11.7	89	36.5

```
In [32]: distributors_2020.tail()
```

```
Out[32]:
```

	Year	Distributor	Market share	Films on release	Box office gross (£ million)
6	2020	Lionsgate	4.5	29	14.0
7	2020	StudioCanal	4.5	46	13.9
8	2020	Entertainment Film Distributors	4.3	6	13.3
9	2020	Shear Entertainment*	1.3	1	4.0
11	2020	Other distributors (143)	7.5	534	23.2

2021

```
In [33]: tables_2021 = extract_tables('./files/2022 - The box office 2021 - full report - 26
```

Reading ./files/2022 - The box office 2021 - full report - 26781.pdf
1 tables found.

```
#####  
      Table 0  
#####
```

	Admissions (million)	2019	2020	2021	% change on 2019
0	January	13.7	16.5	-	-100.0
1	February	12.2	14.5	-	-100.0
2	March	11.4	4.8	-	-100.0
3	April	16.0	-	-	-100.0
4	May	16.6	-	3.5	-78.8
5	June	13.9	-	7.0	-49.9
6	July	18.7	0.4	7.8	-58.5
7	August	15.6	2.1	10.4	-33.1
8	September	11.0	2.7	6.5	-40.6
9	October	16.3	2.2	16.4	0.9
10	November	12.2	0.3	8.8	-27.5
11	December	18.5	0.5	13.5	-26.8

```
In [34]: admissions_2021 = (  
    tables_2021[0]  
    .iloc[:,[0, 3]]  
)
```

```
admissions_2021.columns = ["Month", "Admissions (million)"]
admissions_2021.insert(0, "Year", 2021)

admissions_2021["Admissions (million)"] = admissions_2021["Admissions (million)"].r
admissions_2021.head()
```

C:\Users\david\AppData\Local\Temp\ipykernel_9920\1718005333.py:9: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.
Try using `.loc[row_indexer,col_indexer] = value` instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
admissions_2021["Admissions (million)"] = admissions_2021["Admissions (million)"].
replace("-", np.NaN)
```

Out[34]:

	Year	Month	Admissions (million)
0	2021	January	NaN
1	2021	February	NaN
2	2021	March	NaN
3	2021	April	NaN
4	2021	May	3.5

In [35]: admissions_2021.tail()

Out[35]:

	Year	Month	Admissions (million)
7	2021	August	10.4
8	2021	September	6.5
9	2021	October	16.4
10	2021	November	8.8
11	2021	December	13.5

Genres

For the first time these stats are actually in another PDF

In [36]: tables_2021_genre = extract_tables("./files/2022 - Top films in 2021 - full report

Reading ./files/2022 - Top films in 2021 - full report - 29763.pdf
1 tables found.

```
#####
      Table 0
#####
```

	Genre	Number\nof releases	% of releases	Box office\ngross\n(£ million)	% of total\ntax office	Average number\nof sites at\nwidest point\nof release
0	Action	52	11.8	296.7	48.2	210
1	Comedy	54	12.2	65.1	10.6	138
2	Animation	38	8.6	58.2	9.4	260
3	Horror	24	5.4	40.8	6.6	277
4	Family	8	1.8	29.6	4.8	187
5	Sci fi	4	0.9	29.6	4.8	420
6	Drama	134	30.3	20.6	3.3	78
7	Musical	8	1.8	15.5	2.5	313
8	Adventure	6	1.4	15.2	2.5	243
9	Fantasy	3	0.7	14.9	2.4	263
10	Biopic	5	1.1	13.4	2.2	345
11	Thriller	10	2.3	8.2	1.3	173
12	Suspense	12	2.7	4.5	0.7	117
13	Documentary	69	15.6	2.1	0.3	22
14	Romance	13	2.9	1.5	0.2	77
15	Crime	2	0.5	0.3	<0.1	137

```
In [37]: genres_2021 = (  
          tables_2021_genre[0]
```

```

    .drop(columns=[tables_2021_genre[0].columns[2], tables_2021_genre[0].columns[4]
)

genres_2021.insert(0, "Year", 2021)

genres_2021.columns = ["Year", "Genre", "Number of releases", "Gross box office (£

# fix data errors
genres_2021["Top performing title"] = genres_2021["Top performing title"].str.repla

genres_2021.head()

```

Out[37]:

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
0	2021	Action	52	296.7	No Time to Die
1	2021	Comedy	54	65.1	Free Guy
2	2021	Animation	38	58.2	The Addams Family 2
3	2021	Horror	24	40.8	A Quiet Place Part II
4	2021	Family	8	29.6	Peter Rabbit 2: The Runaway

In [38]: genres_2021.tail()

Out[38]:

	Year	Genre	Number of releases	Gross box office (£ million)	Top performing title
11	2021	Thriller	10	8.2	The Forever Purge
12	2021	Suspense	12	4.5	Old
13	2021	Documentary	69	2.1	Summer of Soul (...Or, When the Revolution Cou...
14	2021	Romance	13	1.5	The Last Letter from Your Lover
15	2021	Crime	2	0.3	Rise of the Footsoldier: Origins

Distributors

In [39]: tables_2021_dist = extract_tables("./files/2022 - Distribution - full report - 2678

Reading ./files/2022 - Distribution - full report - 26782.pdf
1 tables found.

```

#####
      Table 0
#####

```

	Distributor	Market share in 2021 (%)	Films on release in 2021	Box office gross in 2021 (£ million)
0	Universal	30.6	51	180.8
1	Sony	22.2	52	130.8
2	Walt Disney	21.2	63	125.1
3	Warner Bros	14.4	80	84.7
4	Paramount	3.7	16	21.8
5	Lionsgate	1.8	23	10.5
6	Entertainment One	1.4	12	8.1
7	Park Circus	0.6	96	3.4
8	STX Entertainment	0.6	3	3.4
9	StudioCanal	0.5	36	2.7
10	Top 10	96.8	432	571.2
11	Other distributors (137)	3.2	549	19.0

```
In [40]: distributors_2021 = (
    tables_2021_dist[0]
    .drop(index=[10]) # drop Top 10 total row
    )

distributors_2021.insert(0, "Year", 2021)
distributors_2021.columns = ["Year", "Distributor", "Market share", "Films on release", "Box office gross (£ million)"]
distributors_2021.head()
```

```
Out[40]:
```

	Year	Distributor	Market share	Films on release	Box office gross (£ million)
0	2021	Universal	30.6	51	180.8
1	2021	Sony	22.2	52	130.8
2	2021	Walt Disney	21.2	63	125.1
3	2021	Warner Bros	14.4	80	84.7
4	2021	Paramount	3.7	16	21.8

```
In [41]: distributors_2021.tail()
```


Out[41]:

	Year	Distributor	Market share	Films on release	Box office gross (£ million)
6	2021	Entertainment One	1.4	12	8.1
7	2021	Park Circus	0.6	96	3.4
8	2021	STX Entertainment	0.6	3	3.4
9	2021	StudioCanal	0.5	36	2.7
11	2021	Other distributors (137)	3.2	549	19.0

Combine everything

```
In [42]: admissions = pd.concat([admissions_2018, admissions_2019, admissions_2020, admissions_2021],
                                ignore_index=True,
                                axis=0)

admissions.head()
```

Out[42]:

	Year	Month	Admissions (million)
0	2018	January	16.2
1	2018	February	16.1
2	2018	March	13.5
3	2018	April	15.5
4	2018	May	13.7

```
In [43]: admissions["Year"].value_counts()
```

Out[43]:

2018	12
2019	12
2020	12
2021	12

Name: Year, dtype: int64

```
In [44]: admissions.to_csv("admissions.csv", index=False)
```

Genres

```
In [45]: genres = pd.concat([genres_2018, genres_2019, genres_2020, genres_2021],
                              ignore_index=True,
                              axis=0)

genres.head()
```


Out[49]:	Year	Distributor	Market share	Films on release	Box office gross (£ million)
0	2018	Walt Disney	23.6	24	325.6
1	2018	Universal	19.5	40	268.5
2	2018	20th Century Fox	14.5	28	199.3
3	2018	Warner Bros	13.9	31	191.4
4	2018	Sony	10.7	33	146.8

In [50]: `distributors.tail()`

Out[50]:	Year	Distributor	Market share	Films on release	Box office gross (£ million)
39	2021	Entertainment One	1.4	12	8.1
40	2021	Park Circus	0.6	96	3.4
41	2021	STX Entertainment	0.6	3	3.4
42	2021	StudioCanal	0.5	36	2.7
43	2021	Other distributors (137)	3.2	549	19.0

In [51]: `distributors["Year"].value_counts()`

Out[51]: 2018 11
2019 11
2020 11
2021 11
Name: Year, dtype: int64

In [52]: `distributors.to_csv("distributors.csv", index=False)`